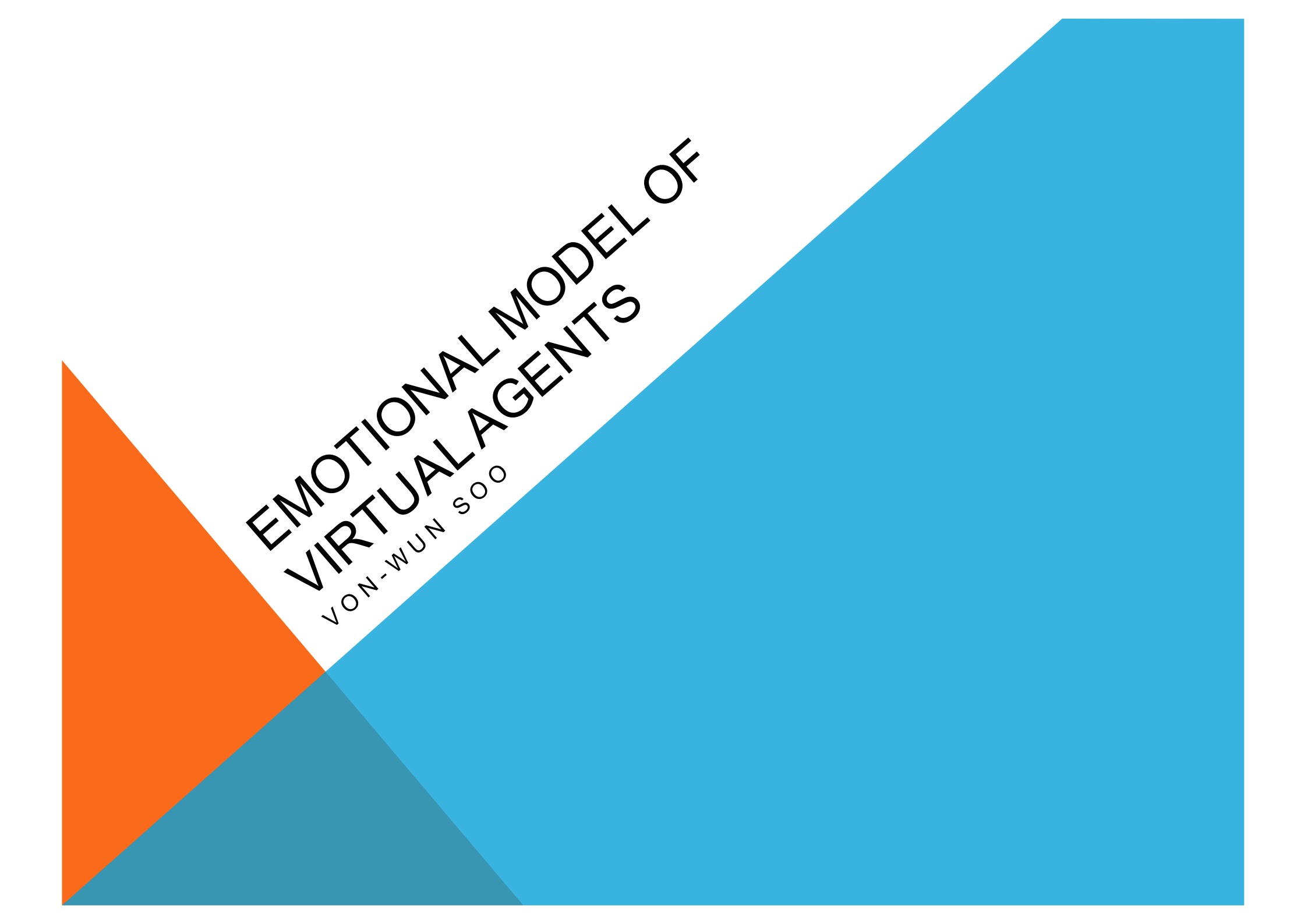




EMOTIONAL MODEL OF VIRTUAL AGENTS

VON-WUN SOO

RATIONAL VIRTUAL AGENTS MODULATED BY EMOTION AND SOCIAL EFFECTS

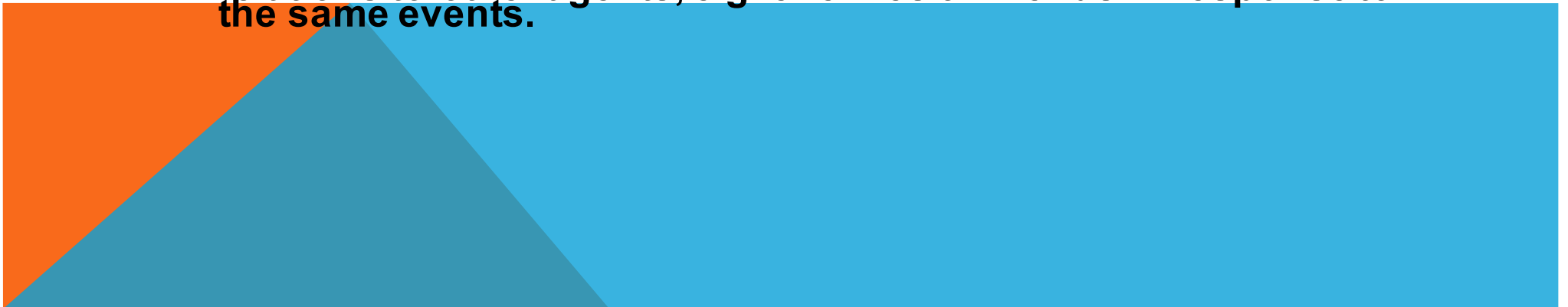
The behaviors of agents are modified by emotions and social effects

The agents might behave “irrationally” e.g. not necessary “good” to themselves.

The agents might behave (express) differently due to different emotional modes or states.

We can infer from facial expressions; speech tones; gestures, etc. via sensors.

The agents might behave differently due to different social relations to other agents, e.g. enemies or friends in response to the same events.



MODELS OF EMOTIONS

Emotion_classification

PAD emotional state model:

(pleasure, arousal, dominance)

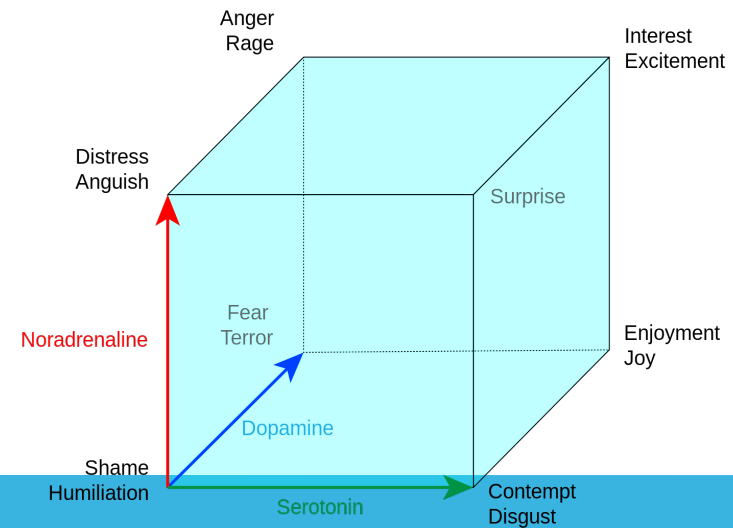
Lövheim Cube of emotion:

serotonin, dopamine, noradrenaline

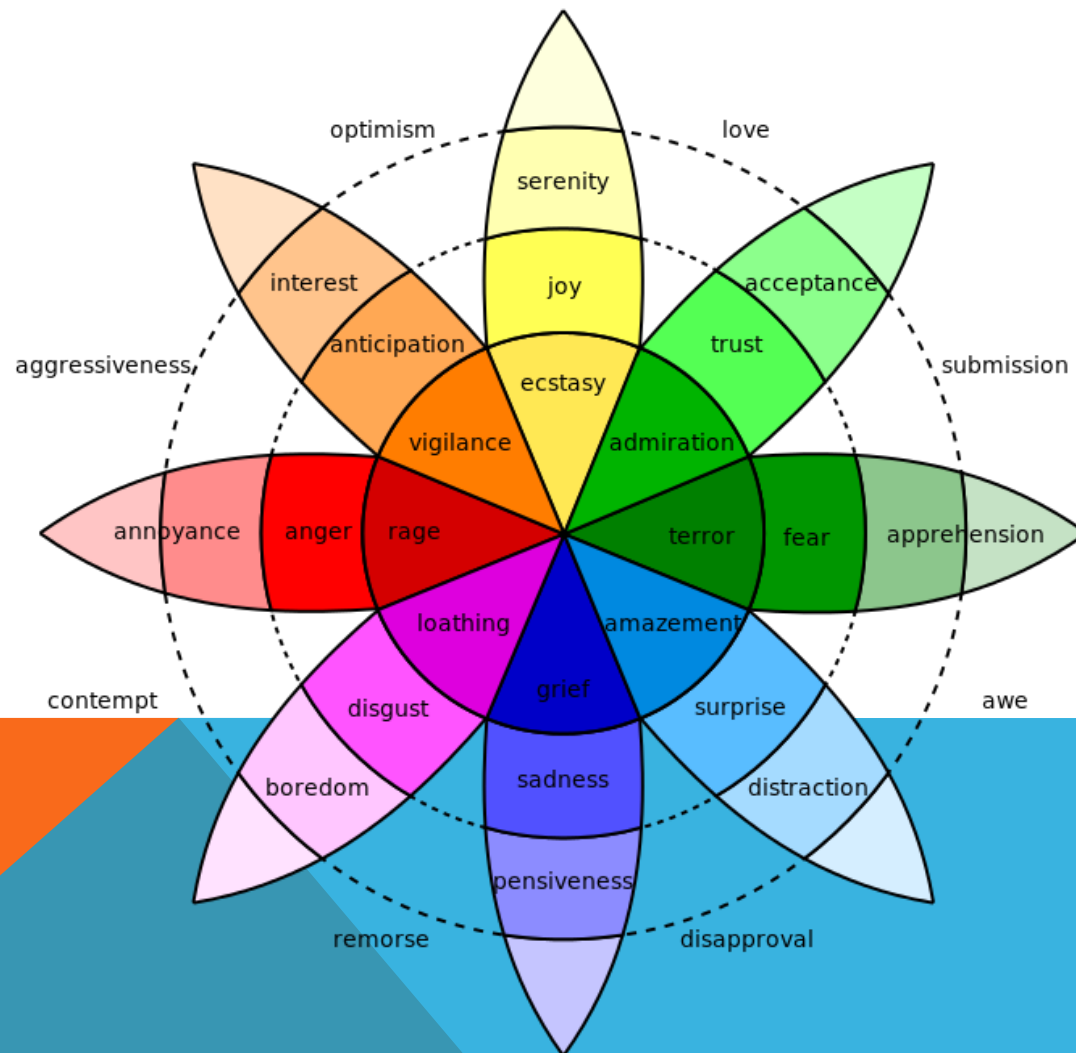
Anger:

high dopamine, noradrenaline

Low serotonin



Robert Plutchik's wheel of emotions



BASIC EMOTION THEORY

Basic emotion types:

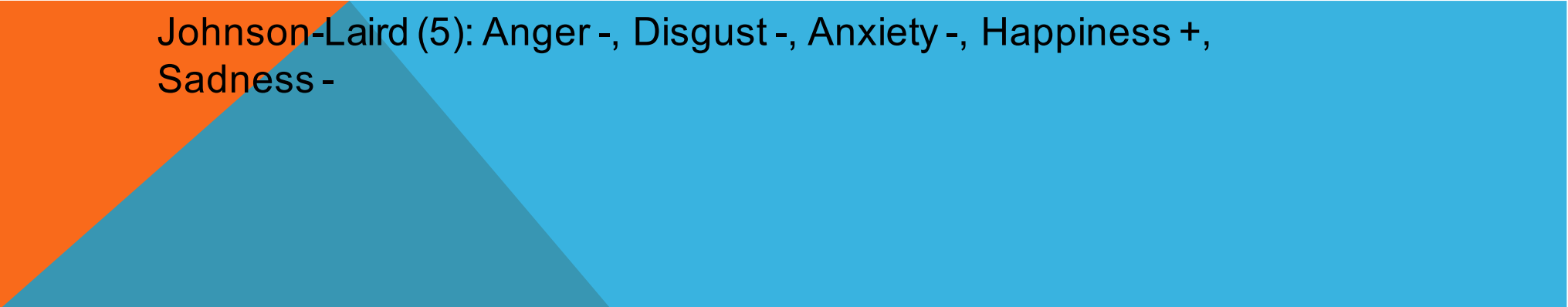
Izard (10): Anger -, Contempt, Disgust -, Distress -, Fear -, Guilt -, Interest +, Joy +, Shame -, Surprise -

Russell's Categories(8): Angry -, Depressed -, Distressed -, Excited +, Miserable -, Pleased +, Relaxed +, Sleepy -

Plutchik (8): Acceptance +, Anger -, Anticipation +, Disgust -, Joy +, Fear -, Sadness -, Surprise -

Ekman(6): Anger -, Disgust -, Fear -, Joy +, Sadness -, Surprise -

Tomkins(9): Anger -, Interest +, Contempt +, Disgust -, Distress -, Fear -, Joy +, Shame -, Surprise -



Johnson-Laird (5): Anger -, Disgust -, Anxiety -, Happiness +, Sadness -

OCC EMOTION MODELS

Ortony, A., Clore, G. L., Collins, A.: The Cognitive Structure of Emotions.
Cambridge University Press (1988)

22 emotion types are modeled and are classified into 3 categories
according to **events, actions, and objects**

+ **surprise and disgust** (**24** emotions)

6 expression groups: joy, sadness, anger, fear, surprise, disgust)

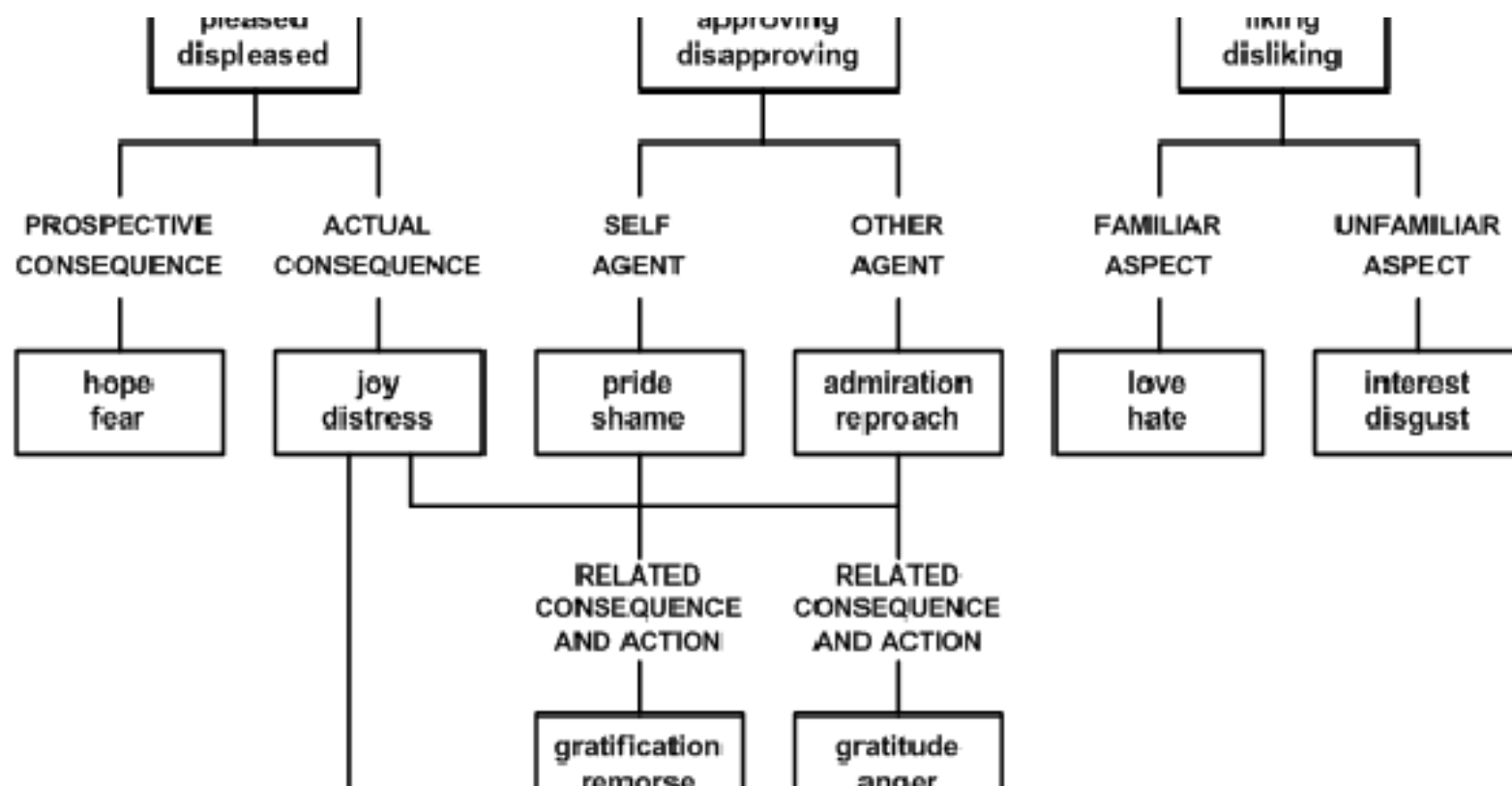


OCC MODELS

Provides an appraisal model to evaluate particular events; actions; or objects perceived from environment wrt goals; norms; attitudes of the agents and their societies.

The appraisal model provides “reasonable” emotional response for those events, actions and objects.





Joy +: (pleased about) a desirable event

Distress -: (displeased about) an undesirable event

Happy-for +: (pleased about) an event presumed to be desirable for someone else

Pity -: (displeased about) an event presumed to be undesirable for someone else

Gloating +; (pleased about) an event presumed to be undesirable for someone else

Resentment -: (displeased about) an event presumed to be desirable for someone else

Hope +: (pleased about) the prospect of a desirable event

Fear -: (displeased about) the prospect of an undesirable event

Satisfaction +: (pleased about) the confirmation of the prospect of a desirable event

Fear-confirmed -: (displeased about) the confirmation of the prospect of an undesirable event

Relief +: (pleased about) the disconfirmation of the prospect of an undesirable event



Disappointment -: (displeased about) the disconfirmation of the prospect of a desirable event

Pride +: (approving of) one's own praiseworthy action

Shame -: (disapproving of) one's own blameworthy action

Admiration+: (approving of) someone else's praiseworthy action

Reproach-: (disapproving of) someone else's blameworthy action

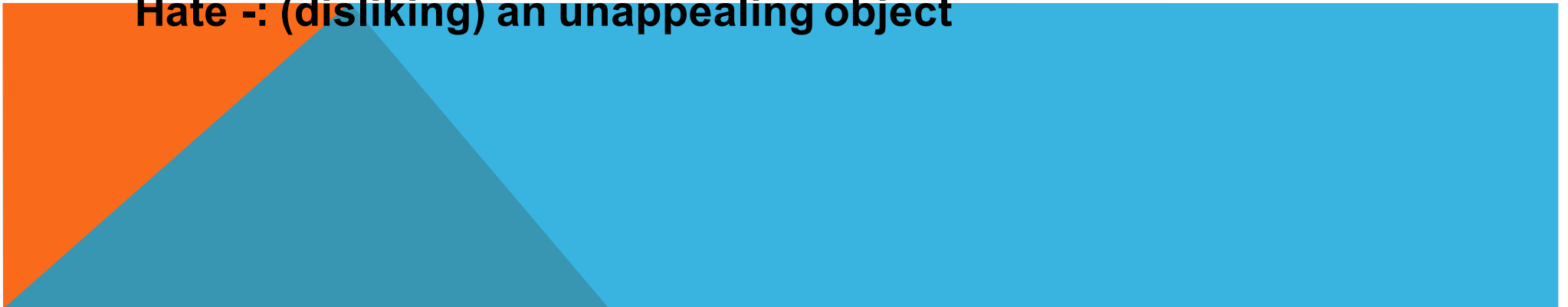
Gratification +: (approving of) one's own praiseworthy action and (being pleased about) the related desirable event (disapproving of) one's Remorse: own blameworthy action and (being displeased about) the related undesirable event

Gratitude + :(approving of) someone else's praiseworthy action and (being pleased about) the related desirable event

Anger -: (disapproving of) someone else's blameworthy action and (being displeased about) the related undesirable event

Love +: (liking) an appealing object

Hate -: (disliking) an unappealing object



RELATIONS OF EMOTIONS TO PERSONALITIES AND SOCIAL RELATIONS

Emotional models would be further modulated by *personality parameters* and social parameters to make precise prediction of emotional response



MOODS

Reflect prolonged state of mind, Cumulated states of emotion

A mood is a relatively lasting affective state. Moods differ from emotions in that they are less specific, often less intense, less likely to be triggered by a particular stimulus or event, however longer lasting.^[1]



MOODS

1Moods generally have either a positive or negative valence. In other words, people often speak of being in a *good* or *bad* mood.

moods generally last for hours or days

Mood also differs from temperament or personality traits which are even more general and long lasting.



PERSONALITY MODELS

Ocean MODEL OF PERSONALITY:

- 5 factors: **Extraversion** (Intraversion), **Agreeableness**, **Conscientiousness** (Lack of conscientiousness), **Neuroticism** (Emotional Stability), and **Openness**(closeness to experiences)

Other model of personality:

- **Optimism; pessimism**



6 FACETS (OR SUBORDINATE TRAITS).

N	E	O	A	C
Anxiety	Warmth	Fantasy	Trust	Competence
Angry hostility	Gregariousness	Aesthetics	Straightforward-ness	Order
Depression	Assertiveness	Feelings	Altruism	Dutifulness
Self-consciousness	Activity	Actions	Compliance	Achievement striving
Impulsiveness	Excitement-seeking	Ideas	Modesty	Self Discipline
Vulnerability	Positive emotion	Values	Tender-mindedness	Deliberation

CRITICISM OF BIG-FIVE FACTOR MODEL

Block (1995) gave a detailed critique of the Big Five in A contrarian view of the five-factor approach to personality description. **Costa and McCrae (1995)** answered this paper in Solid ground in the wetlands of personality: A reply to Block.



One common criticism is that the Big Five does not explain all of human personality.

It ignores such as Religiosity, Manipulativeness/ Machiavellianism, Honesty, Thriftiness, Conservativeness, Masculinity/Femininity, Snobbishness, Sense of humour, Identity, Self-concept, and Motivation.



**McAdams (1995) called the Big Five a "psychology of the stranger,"
because they refer to traits that are relatively easy to observe in a
stranger**

Personality can be privately held or context-dependent



RELATIONS BETWEEN OCEAN AND OCC

	Goals	Standards	Attitudes
Openness		making a shift of standards in new situations	attitude towards new elements
Conscientiousness	abandoning and adopting goals, determination		
Extraversion			willingness to communicate
Agreeableness	abandoning and adopting goals in favor of others	compromising standards in favor of others	adaptiveness to other people
Neuroticism			

Table 1. Relationship between OCEAN and OCC parameters.

PERSONALITY TRAIT INVENTORY

<http://darkwing.uoregon.edu/%7Eegsaucier/gsau4.htm>

Mini-markers: scale 1~9 inaccurate~accurate

40 items (lexicons) to describe yourself:



40 FACTORS

____ Bashful ____ Bold ____ Careless ____ Cold ____ Complex
____ Cooperative ____ Creative ____ Deep ____ Disorganized ____ Efficient
____ Energetic ____ Envious ____ Extraverted ____ Fretful ____ Harsh ____
Imaginative ____ Inefficient ____ Intellectual ____ Jealous ____ Kind ____
Moody ____ Organized ____ Philosophical ____ Practical ____ Quiet
____ Relaxed ____ Rude ____ Shy ____ Sloppy ____ Sympathetic
____ Systematic ____ Talkative ____ Temperamental ____ Touchy ____ Uncreative
____ Unenvious ____ Unintellectual ____ Unsympathetic ____ Warm ____
Withdrawn



16 PERSONALITY FACTOR QUESTIONNAIRE

16 personality 16PF



MYERS-BRIGGS TYPE INDICATOR

Type not trait.

No right or wrong

Popular in business

But poor reliability

Myers-Briggs_Type_Indicator



16 TYPES

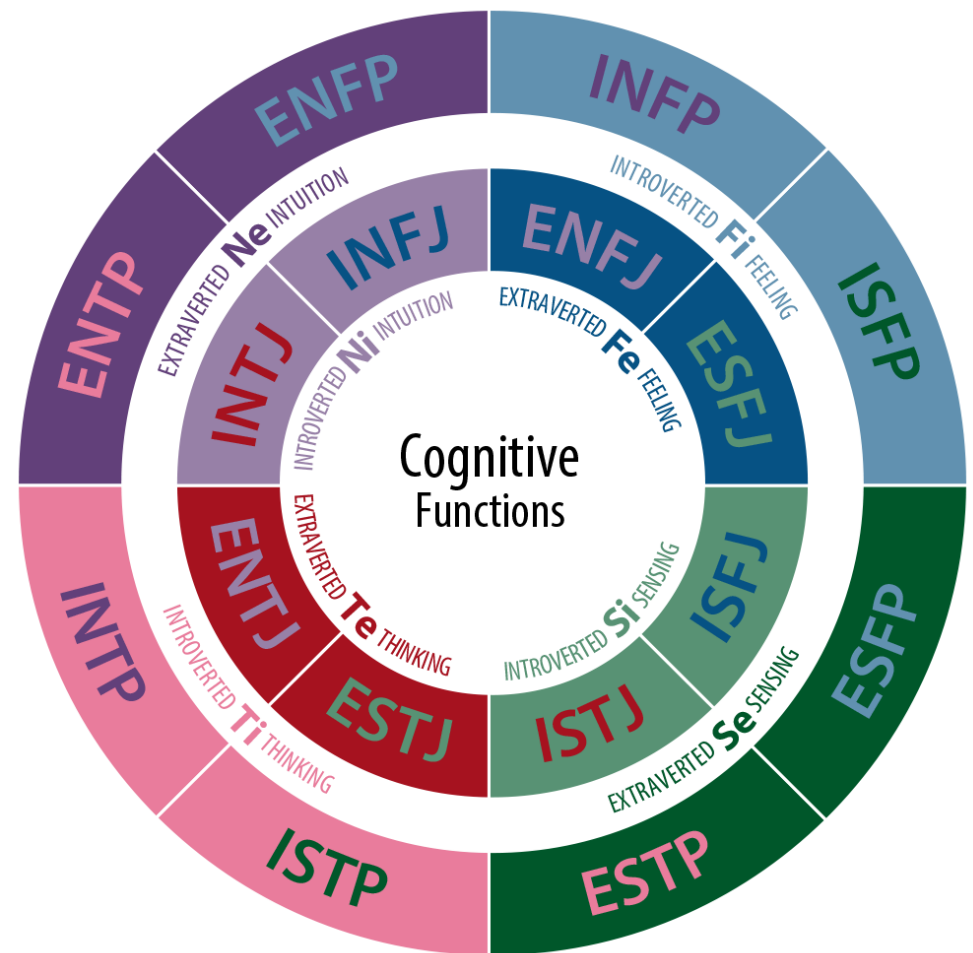
Extraversion/Introversion

Sensing/Intuition

Thinking/Feeling

Judging/Perceiving

E,I,S,I,T,F,J,P



IS_IT_EJ	IS_IF_EJ	IN_IF_EJ	IN_IT_EJ
Inspector	Protector	Counselor	Mastermind
IS_ET_IP	IS_EF_IP	IN_EF_IP	IN_ET_IP
Crafter	Composer	Healer	Architect
ES_ET_IP	ES_EF_IP	EN_EF_IP	EN_ET_IP
Promoter	Performer	Champion	Inventor
ES_IT_EJ	ES_IF_EJ	EN_IF_EJ	EN_IT_EJ
Supervisor	Provider	Teacher	Fieldmarshal



MBTI MODEL CORRELATION WITH BIG-FIVE MODEL

MBTI lacks of emotion stability dimension

	Extraversion		Openness		Agreeableness
	Conscientiousness		Neuroticism		
E-I	-0.74	0.03	-0.03	0.08	0.16
S-N	0.10	0.72	0.04	-0.15	-0.06
T-F	0.19	0.02	0.44	-0.15	0.06
J-P	0.15	0.30	-0.06	-0.49	0.11

The closer the number is to 1.0 or -1.0, the higher the degree of correlation.



EXAMPLE OF PERSONALITY INTERACTING WITH EMOTION

Julie is standing outside. She has to carry a heavy box upstairs. A passing man offers to help her carry the box upstairs. Julies personality has a big influence on her perception and on her behaviour. If she has an extravert personality, she will be happy that someone offers her some help. If she has a highly introvert and/or neurotic personality, she will feel fear and distress and she will respond differently.



Julie's behaviour as an extravert (a) and an introvert (b) personality.

<Man> Can I help you carry this box upstairs?
<Julie> |GR30| Oh Thank you!
That is so nice of you to ask!



<Man> On what floor are you living?
<Julie> |GR99|The third floor. It's quite high!
I am so glad you are helping me out!

(a)

<Man> Can I help you carry this box upstairs?
<Julie> |FE34|No I don't need your help.



<Man> On what floor are you living?
<Julie> |FE99|That is none of your business.
Go away!

(b)

REFERENCES

A Model for Personality and Emotion Simulation

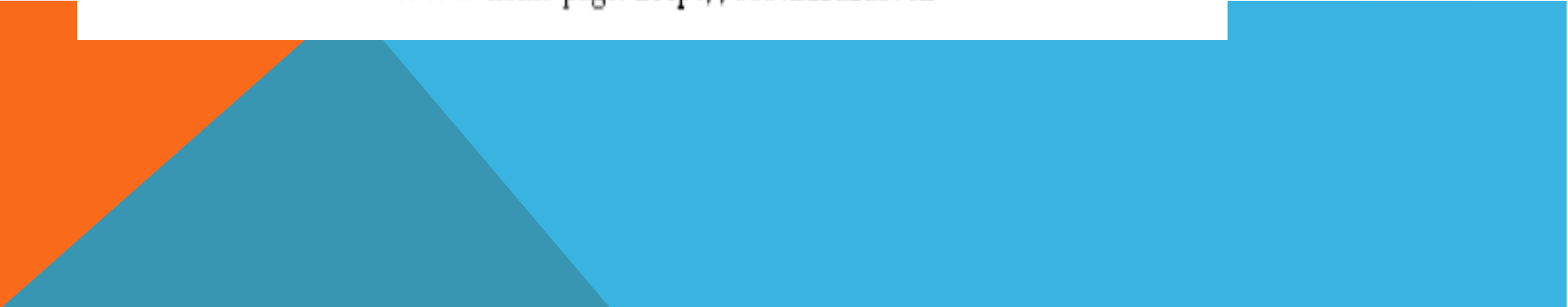
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